

Lecture 3: Universal Properties

Summary.

- (1) Universal properties: concept, examples and ubiquity.
- (2) Initial and final objects in a category.
- (3) Universal characterisation of Cartesian product in $\mathbf{Set}$. The categorical product construction.
- (4) Universal properties ‘come in pairs’: the coproduct construction. Properties of products and coproducts.

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Opening.

If there is a ‘main topic’ in category theory, this is certainly the study of *universal* properties. Roughly speaking, an entity ϵ is universal among a family of ‘similar’ entities if it is the case that every other entity in the family can be *reduced* or *traced back* to ϵ . For example, an object T is said to be *final* in a category C if, from every other object X in C , there exists a unique arrow $!_X$ to T . Therefore, there is a canonical, *unique* way to relate every object in C to T — *finality* is thus an universal property.

A nice thing about universal properties is the fact they always ‘come in pairs’: the *dual* of an universal is still an universal. Dualising finality, we arrive at *initiality*: an object is *initial* in C if there is one and only one arrow in C from it to any other object in the category.

Universal properties, like finality or initiality, can be recognised, usually under a different terminology, in many branches of Mathematics. Moreover, they happen to play a major role in the structure of ‘mathematical spaces’. Therefore, category theory provides a setting for studying abstractly such ‘spaces’ and their relationships.

Let us consider an illustrative example (adapted from [2]). The study of bilinear (i.e. linear in both arguments) maps out of two vector spaces U and V can be reduced to the study of linear maps because there is a *universal* bilinear map $\epsilon : U \times V \longrightarrow T$ through which all the others factor, i.e. for all $f : U \times V \longrightarrow X$, there exists one and only one linear map $\bar{f} : T \longrightarrow X$ such that $f = \bar{f} \cdot \epsilon$. Look for a moment how uniqueness is proved. Suppose both ϵ and $\epsilon' : U \times V \longrightarrow T'$ satisfy the universal property above. Thus, we obtain linear maps $\bar{\epsilon}$ and $\bar{\epsilon}'$, such that

$$\epsilon' = \bar{\epsilon}' \cdot \epsilon \quad \text{and} \quad \epsilon = \bar{\epsilon} \cdot \epsilon'$$

because, respectively, ϵ and ϵ' are universal by assumption. Clearly, $\bar{\epsilon} \cdot \bar{\epsilon}' \cdot \epsilon = \bar{\epsilon} \cdot \epsilon' = \epsilon$ as depicted in the following diagram:

$$\begin{array}{ccccc} & & U \times V & & \\ & \swarrow \epsilon & \downarrow \epsilon' & \searrow \epsilon & \\ T & \xrightarrow{\bar{\epsilon}'} & T' & \xrightarrow{\bar{\epsilon}} & T \end{array}$$

However, $\text{id}_T \cdot \epsilon = \epsilon$, which entails $\bar{\epsilon} \cdot \bar{\epsilon}' = \text{id}_T$ by the uniqueness of ϵ . A similar argument, relying on the universality of ϵ' , yields $\bar{\epsilon}' \cdot \bar{\epsilon} = \text{id}_{T'}$. Thus, $\bar{\epsilon}$ is an isomorphism witnessing $T \cong T'$.

Vector space T is the *tensor* product of U and V , often written as $U \otimes V$; and what the universal property tells is that it is essentially unique. The way it is constructed is, to a large extent, irrelevant: the universal property is enough.

Exercise 1

Characterise the initial and final objects in a preorder regarded as a category. Give an example of a preorder in which such objects do not exist.

Exercise 2

Show that any singleton set is both initial and final in $\text{Set}_\perp$ (and, therefore, called a *zero* object). Can you think of another familiar category with a zero object?

Exercise 3

Let Rng be the category of rings and consider $Z = \langle \mathbb{Z}, +, 0, -, \cdot, 1 \rangle$ the ring of integer numbers. Show that there is a unique ring homomorphism h from Z to any other ring $\langle S, +', 0', -, \cdot', 1' \rangle$ given by

$$h(n) \triangleq \begin{cases} 0' & \Leftarrow n = 0 \\ -'h(-n) & \Leftarrow n < 0 \\ \underbrace{1' + ' 1' + ' \dots + ' 1'}_n & \Leftarrow n > 0 \end{cases}$$

Exercise 4

Based on the previous exercise, conclude that Z is the initial object in Rng , showing that any other ring satisfying the universal property is isomorphic to Z .

Exercise 5

Show that any map from a final object in a category to an initial one is an isomorphism.

Exercise 6

Coalgebras are a generic way represent transition systems. Formally, a *coalgebra* for a functor $F : C \rightarrow C$, thought of as the type of the allowed transitions, is an object U , called its carrier, or state space, and an arrow $c : U \rightarrow F(U)$ of C . A morphism between coalgebras c and c' is an arrow $h : U \rightarrow V$ in C making the following diagram commute:

$$\begin{array}{ccc} U & \xrightarrow{h} & V \\ c \downarrow & & \downarrow c' \\ F(U) & \xrightarrow{F(h)} & F(V) \end{array}$$

1. Instantiate the definition for $C = \text{Set}$ and $F(X) = \mathcal{P}(L \times X)$, where $\mathcal{P}$ is the finite powerfunctor and L an arbitrary set (of labels, say). What sort of transition systems correspond to this type of coalgebras?
 2. Show that coalgebras and their morphisms form a category.
 3. Prove that, if coalgebra $(W, \omega : W \rightarrow F(W))$ is final in the category of F -coalgebras, ω is an isomorphism.
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Exercise 7

Dualise the definition of a coalgebra given above to arrive to the dual concept of a *F-algebra*, $(A, \alpha : F(A) \rightarrow A)$. Show that an initial algebra in the corresponding category is also an isomorphism — notice the proof structure is exactly the same used in the last question of the previous exercise.

Exercise 8

Characterise product and coproduct in a poset regarded as a category. Do the same for the category Pos whose objects are posets and arrows are monotone functions.

Exercise 9

Characterise product and coproduct in a discrete category.

Exercise 10

Resorting to the corresponding universal property, show that the product (respectively, coproduct) construction in a category is functorial. Show, in particular that, given two arrows $f : A \rightarrow B$ and $g : C \rightarrow D$, $f \times g : A \times C \rightarrow B \times D = \langle f \cdot \pi_1, g \cdot \pi_2 \rangle$. What about $f + g$?

Exercise 11

Derive, from the universal property of products, the equality $\langle f, g \rangle \cdot h = \langle f \cdot h, g \cdot h \rangle$, for f, g and h suitably typed, and $\langle \text{id}_A, \text{id}_B \rangle = \text{id}_{A \times B}$. These results are known in classical program calculi [1], as the product *fusion* and *reflection* laws, respectively.

Exercise 12

A coproduct in Rel is given by disjoint union, with the universal arrow in the diagram below defined as

$$[R, S] \triangleq R \cdot \iota_1^\circ \cup S \cdot \iota_2^\circ$$

$$\begin{array}{ccccc} A & \xrightarrow{\iota_1} & A + B & \xleftarrow{\iota_2} & B \\ & \searrow R & \downarrow [R, S] & \swarrow S & \\ & & C & & \end{array}$$

Define product in Rel by dualising this construction. Recall that Rel is a self-dual category.

Exercise 13

The product of two vector spaces U, V over a field K , in Vect_K usually represented as $U \oplus V$, is given by $U \times V = \{(u, v) \mid u \in U, v \in V\}$ made into a vector space by defining addition and scalar multiplication as follows:

$$(x, y) + (x', y') = (x + x', y + y') \text{ and } k(x, y) = (kx, ky)$$

Projections and the universal arrow are as in Set but required to be linear. Show such is the case indeed.

Exercise 14

The Cartesian product $U \oplus V$ of two vector spaces U, V over a field K , in Vect_K , is simultaneously their product (as discussed in the previous exercise) and coproduct. Define the embeddings $\iota_1 : U \rightarrow U \oplus V$ and $\iota_2 : V \rightarrow U \oplus V$ as

$$\iota_1(x) = (x, 0_V) \text{ and } \iota_2(y) = (0_U, y)$$

where $0_U, 0_V$ are the additive identities in U and V , respectively. For $f : U \rightarrow Z$ and $g : V \rightarrow Z$, define the universal arrow $[f, g] : U \oplus V \rightarrow Z$ by

$$[f, g](x, y) = f(x) + g(y)$$

and prove that the relevant arrows are linear and this construction defines indeed a coproduct.

References

- [1] R. Bird and O. Moor. *The Algebra of Programming*. Series in Computer Science. Prentice-Hall International, 1997.
- [2] T. Leinster. *Basic Category Theory*. Cambridge Studies in Advanced Mathematics. Cambridge University Press, 2014.